When AI meets reality

Build the Brain Behind the Machine

HackUPC 2026

Vision and context

Vision:

The future of manufacturing is about smarter collaboration between humans and machines.

Your mission is to **build a Digital Co-Pilot: an intelligent partner** that understands the machine's physics, predicts its behavior, and empowers operators to act with confidence.

Industrial 3D Printer intro & context

Technical Constrains & Rules

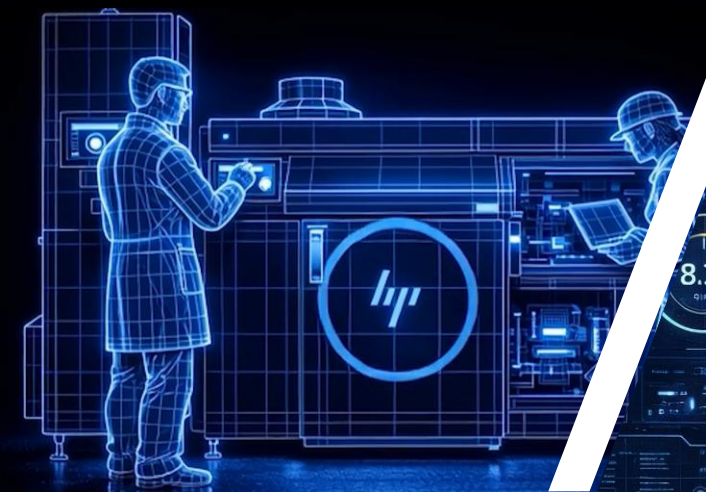
- Free technology stack: choose language and tech
- No pre-made telemetry or historical data needed
- Any AI or LLM components used **will be highly valued,** you will have a choice to include AI tools and features in each step

Challenge phases

Model

Buil a digital replica of a real machine:

Physics meets AI – build a model that responds to the world around it.



Simulate

Run the model and predict failures beforehand

Interact

Build a natural language interface

Challenge phases: Model

Buil a digital replica of a real machine:

Physics meets AI – build a model that responds to the world around it.

Mission: Build the brain of the printer.

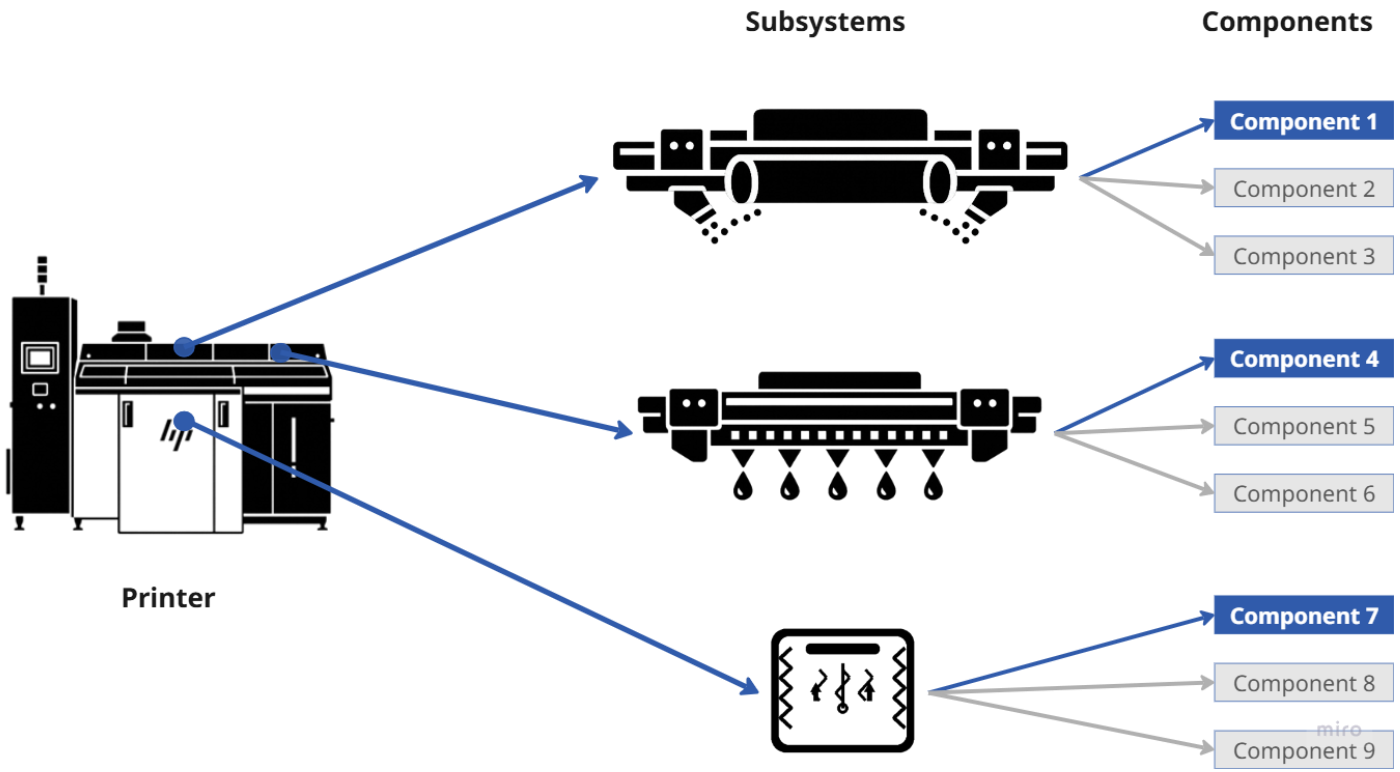
Goal:

- Create an engine that models how components age, fail, and interact under environmental and operational stress.
- This engine ingests conditions as input and produces the state of every component for the digital replica.

Phase 1: Model

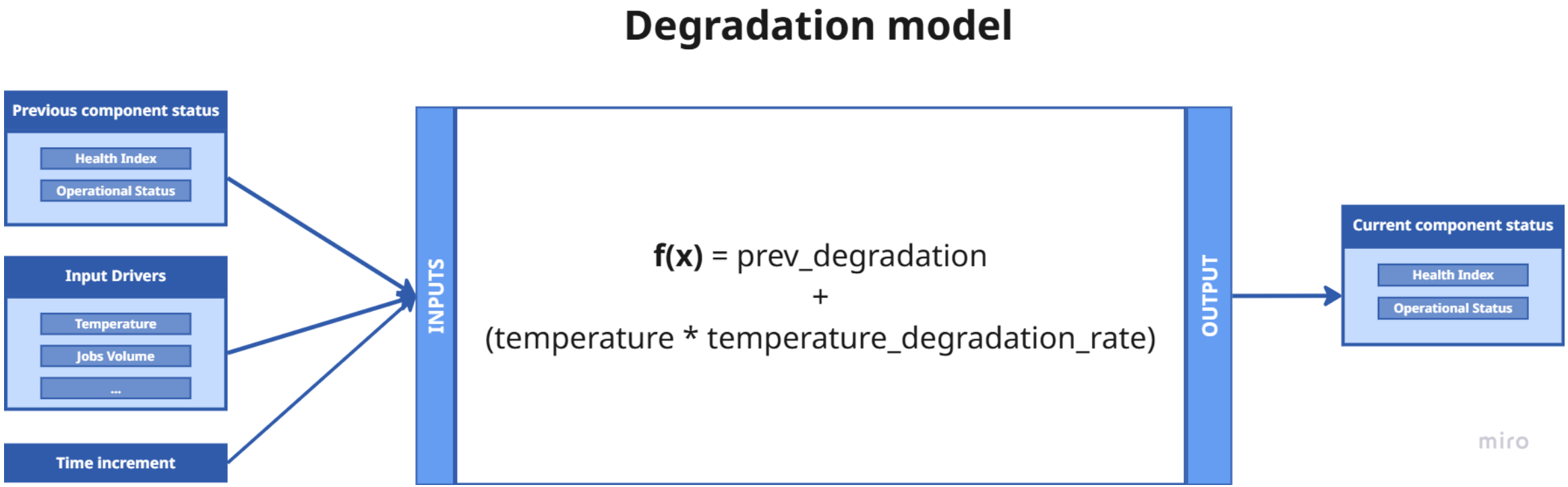
How we structure the model?

- We decompose the printer into **three core subsystems** to simplify modeling.
- Each subsystem must include **at least one** modeled component.
- The structure keeps the model **modular, balanced, and extensible**.



Phase 1: Model

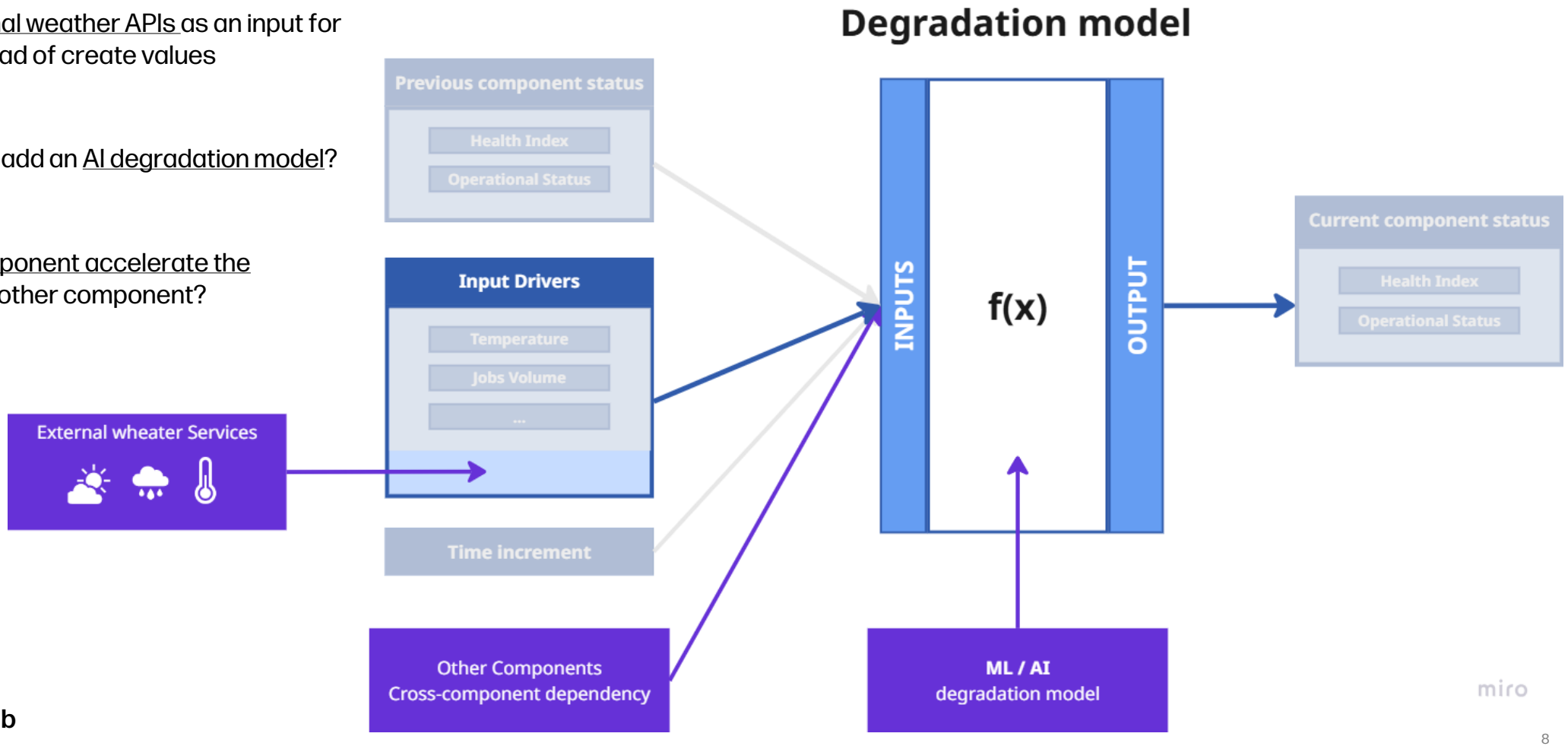
How to create a degradation model?



Phase 1: Model

Go further and show what you can do

- **Idea 1**: Use external weather APIs as an input for input drivers Instead of create values
- **Idea 2**: What if we add an AI degradation model?
- **Idea 3**: Can a component accelerate the degradation of another component?



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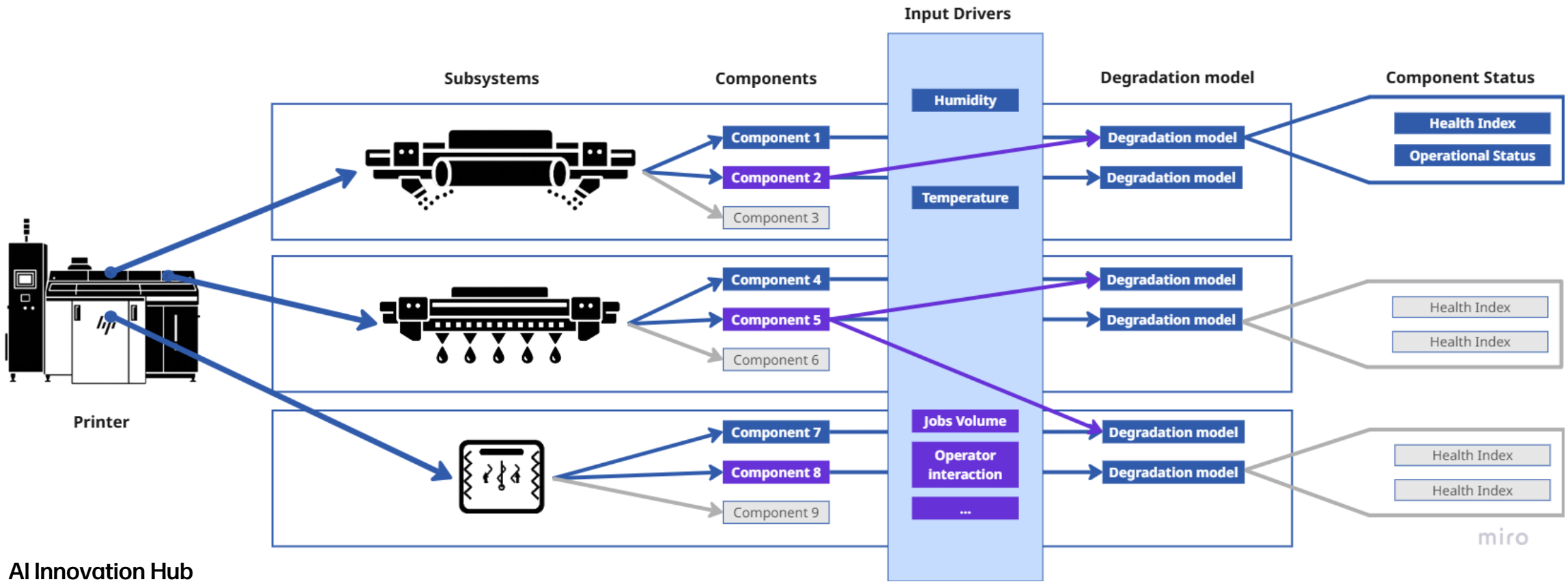
Phase 1: Model

Go further and show what you can do

Idea 3: Can a component accelerate the degradation of another component?

Idea 4: Multiple input drivers per degradation model?



Phase 1: Model – Output and evaluation

Deliverables:

- **Minimum:** 3 degradation models coded that use at least 1 input driver
- **Explanation** of how the degradation models have been implemented

Evaluation Criteria:

- **Rigor**: Logically consistent degradation and failure models
- **Systemic Interaction**: realistic component reactions to environmental and operational drivers
- **Complexity & Innovation (Bonus):** Advanced behaviors such as cascading failures or stochastic models: **surprise us!**
- **Realism & Fidelity (Bonus):** Accuracy in reflecting real-world system behavior

Challenge phases: Simulation

Run a simulation of how parameters evolve over time

Mission: Transform your Stage 1 Modeling into a dynamic, time-advancing digital model of the machine

Goal:

- Build the Simulation Engine (“The Clock”).
- Simulate input drivers over time (i.e. how does temperature evolve?).
- Timestamped history (and future!) of the printer's health and operational state.

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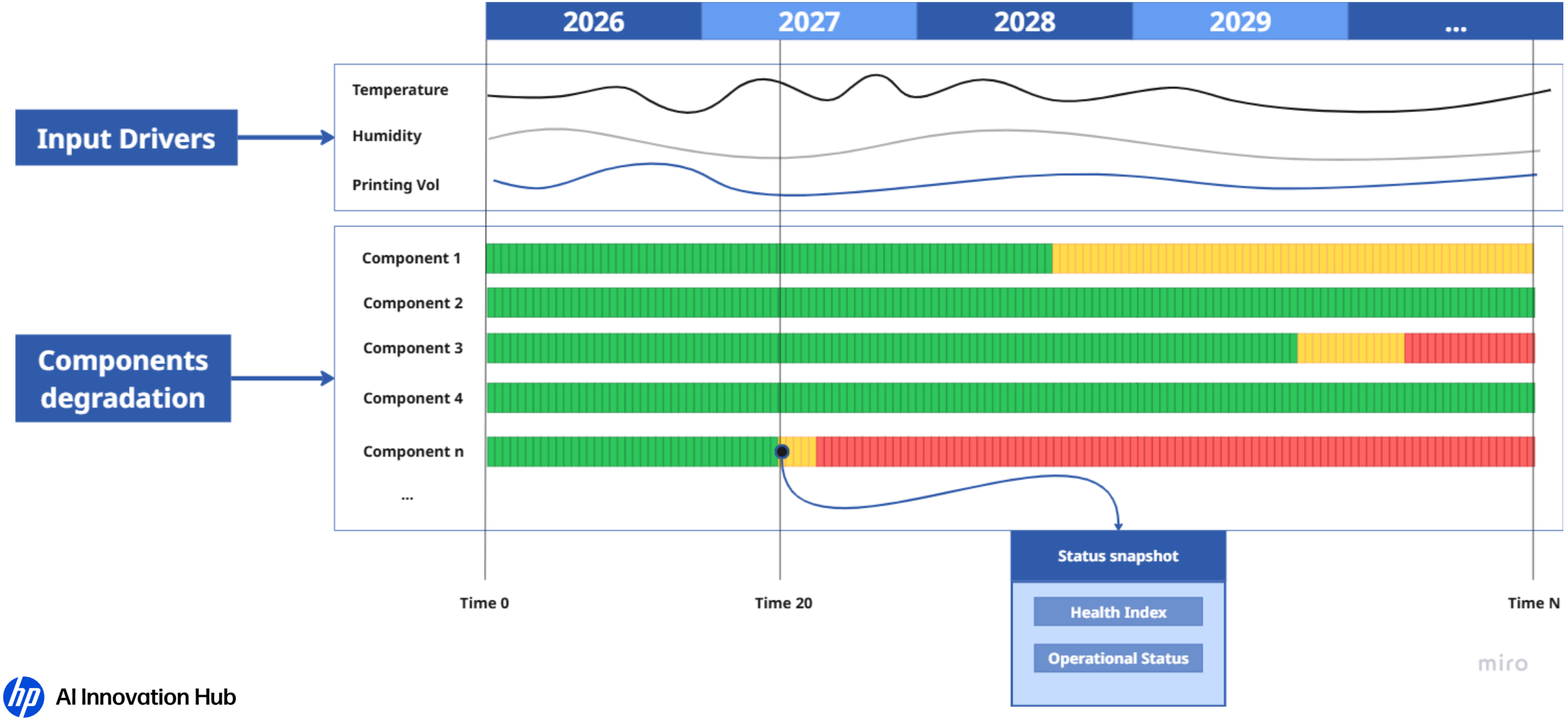
Phase 2: Simulate



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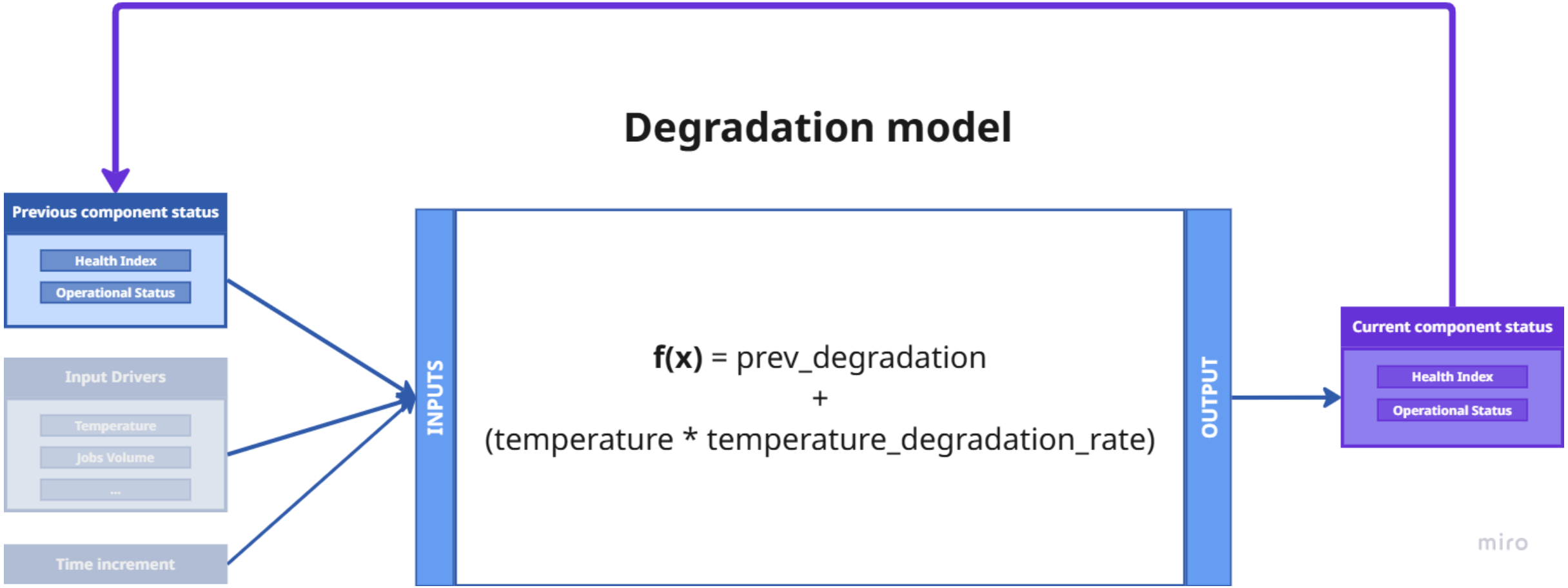
Phase 2: Simulate



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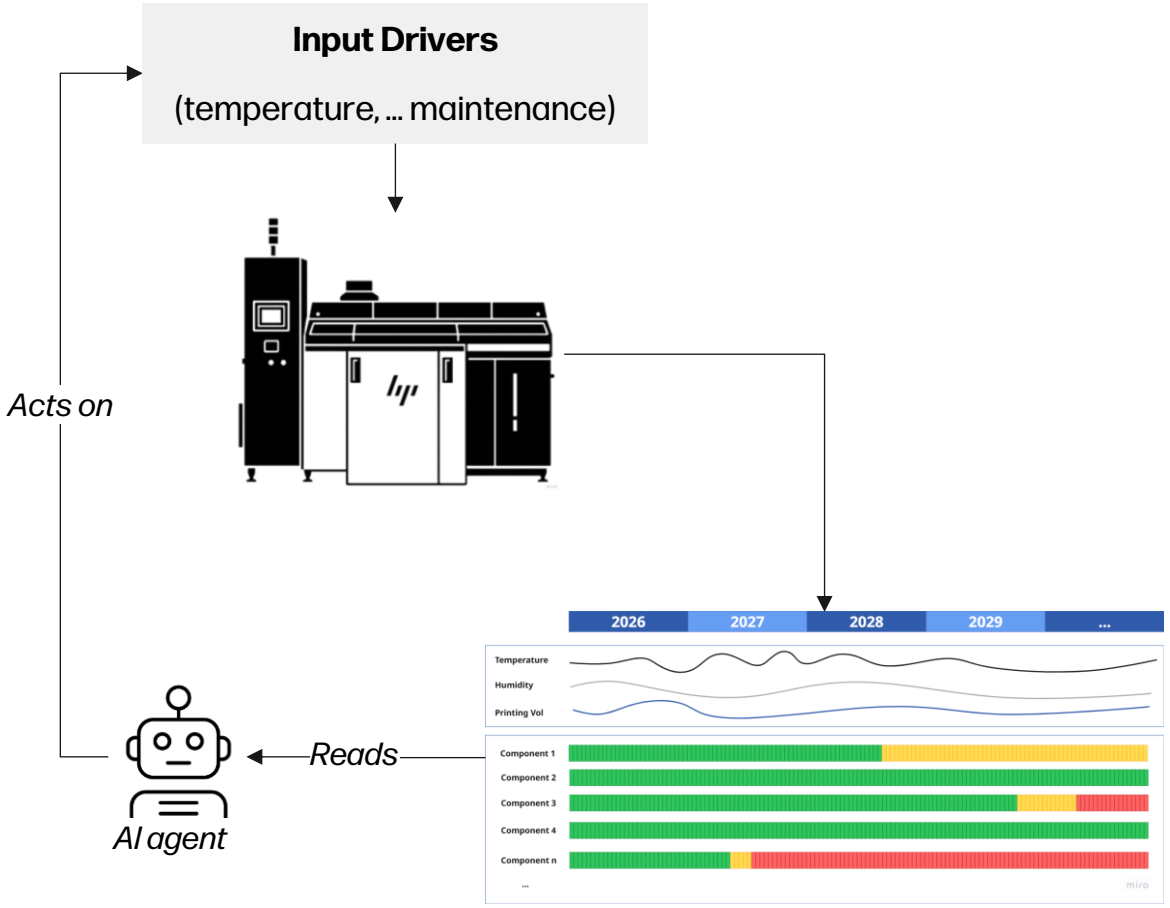
Phase 2: Simulate

Idea 1: Your digital model doesn't just record history. You build an agent that autonomously decides when to trigger maintenance to optimize

Idea 2: Reinforcement Learning Agent: learn the optimal maintenance policy by interacting with the simulation over thousands of episodes

Idea 3: Digital Twin Synchronization: adjust the digital predicted state against incoming "real" sensor readings.



Phase 2: Simulate – Output and evaluation

Deliverables:

- **Time Series Visualization:** A running plot or dashboard showing component health evolving over simulated time
- **Degradation Model:** A clear explanation of how each component's decay logic was designed and implemented
- **Failure Analysis:** Identify when each component fails and why

Evaluation Criteria:

- **Time moves**: Simulation loop advances time correctly and consistently.
- **Systemic Interaction**: Stage 1 engine is called at every time step, and results are persisted to a time-series record
- **Complexity & Innovation (Bonus):** Advanced behaviors such as an *AI Maintenance Agent* that autonomously schedules maintenance.... **surprise us**!
- **Realism & Fidelity (Bonus)**: Degradation curves reflect physically plausible behavior, non-linear component interactions are modeled, and every failure event is traceable to its root input drivers

Challenge phases: Interact

Build a natural language interface

Mission: build an intelligent interface that bridges the gap between complex industrial telemetry and human understanding

Goal:

- Transform the raw, timestamped data produced in Stage 2 into actionable intelligence through natural language interaction.
- Showcase your best work in human-machine and AI user experience: Text, voice, or whatever you can imagine.

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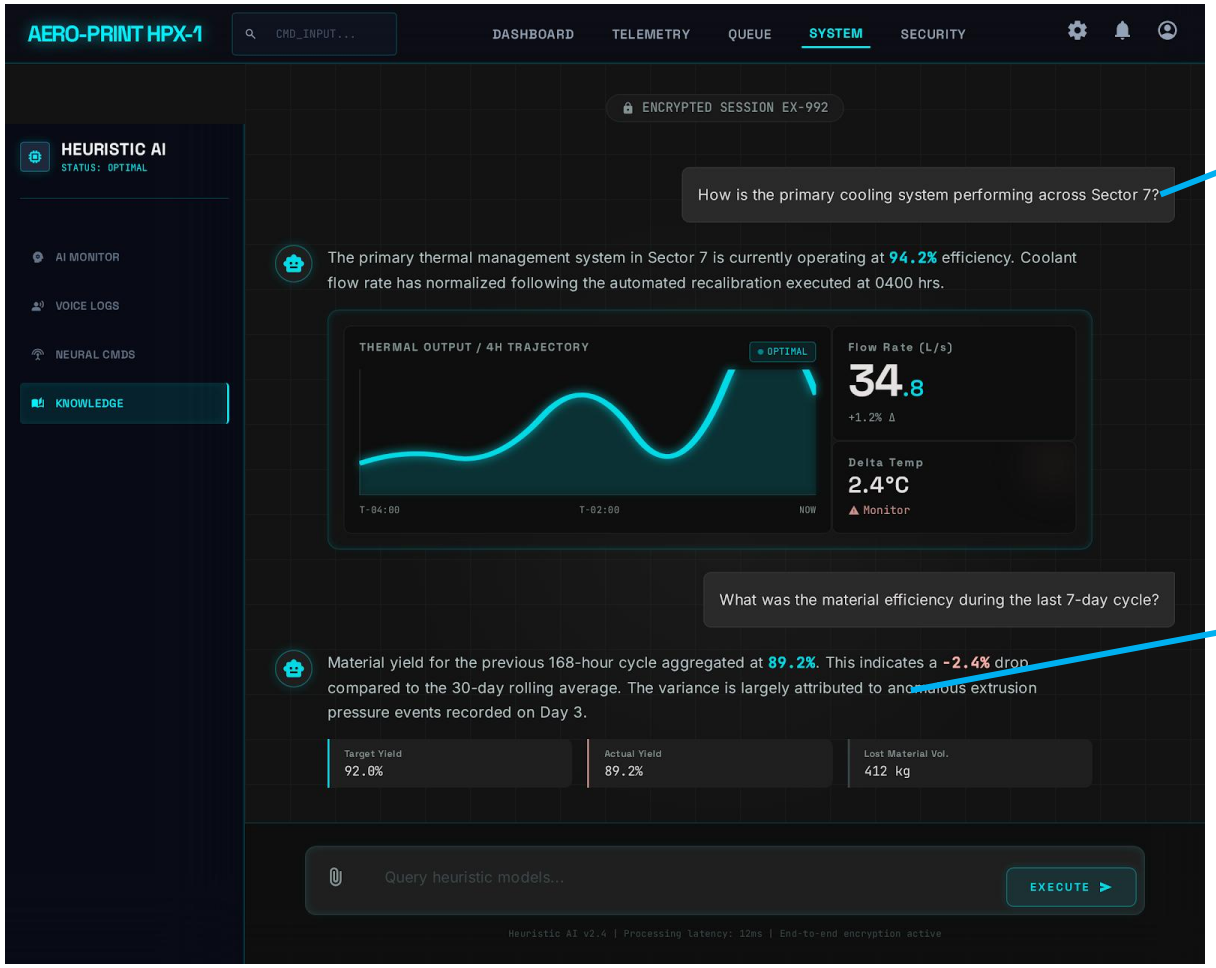
Phase 3: Interact



Provide natural language access to the data

Ground your answers on the generated data

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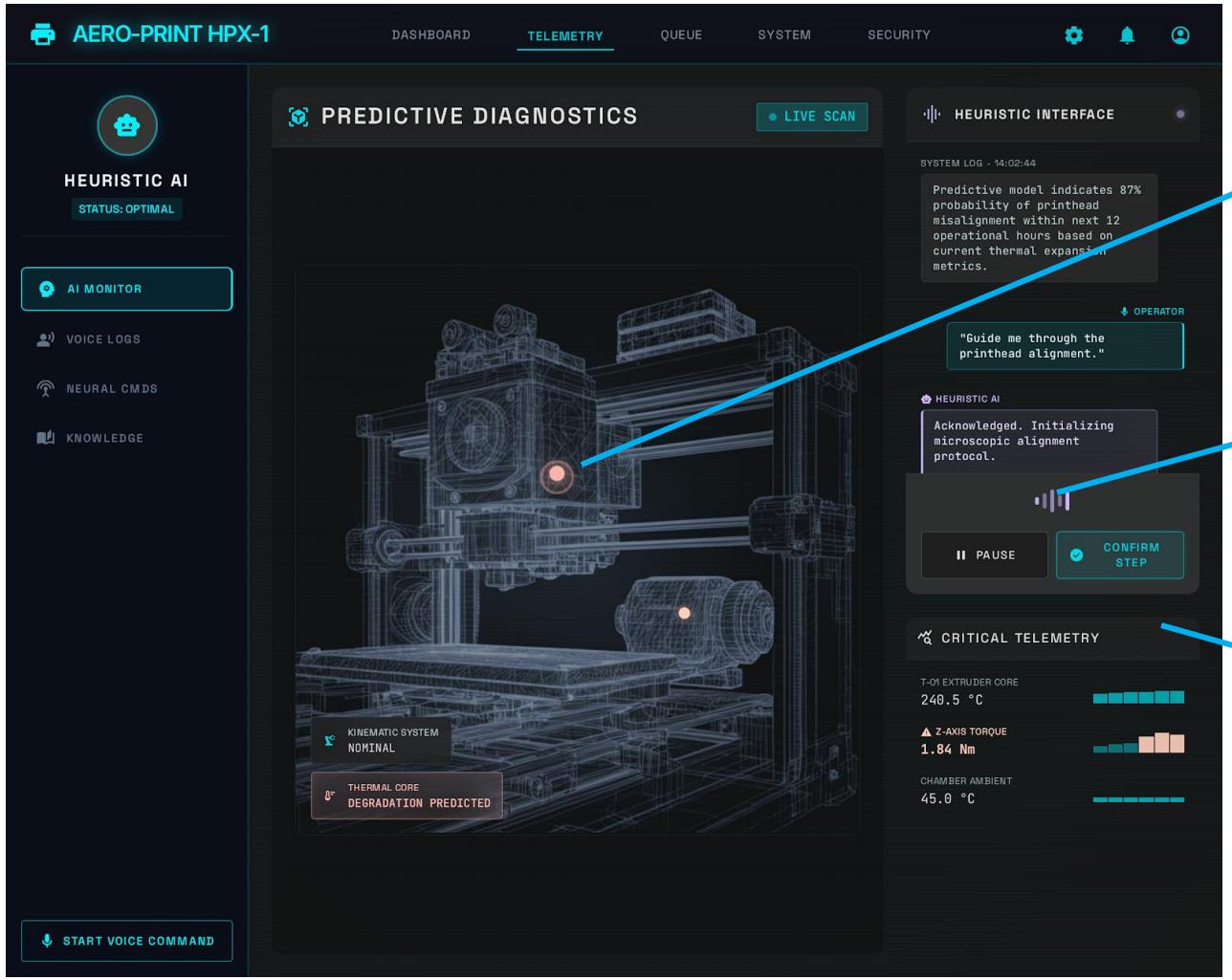
Phase 3: Interact



- Identify failing parts and why
- Allow voice, hands-free, access to the information
- Provide action paths for the user

Phase 3: Interact – Output and evaluation

Deliverables:

- An interface that implements data retrieval reasoning, strictly follows the grounding protocol with no hallucinations, and provides explicit evidence citations for every response.

Evaluation Criteria:

- **Reliability:** Accurate grounding in system data and clear, human-readable communication
- **Intelligence:** Depth of reasoning beyond retrieval, including diagnostic analysis
- **Autonomy:** Proactive behavior and intelligent assistance without user prompts
- **Versatility:** Effective interaction modes and advanced agent capabilities (bonus paths)

Supporting materials

- **Complete Markdown Documentation** provided for the full challenge
- **Clear context and vision** to understand the problem you're solving
- **Detailed breakdown of each phase**, objectives, and deliverables
- **Practical hints and guidance** to help you move past blockers
- **Optional extensions and ideas** to explore deeper, advanced concepts

Designed so you can spend less time guessing—and more time building the cool parts.

Plus, a team of HP volunteers that will be here during all the weekend to help, coach and guide you!

🏆 Hackathon Brief: Digital Twinning a High-Tech 3D Printer

1. The Vision

> The future of manufacturing is not just about more powerful machines;

As industrial hardware like the **HP Metal Jet S100** reaches unpreceden
wider. An operator can no longer rely on manual monitoring or reactive
dashboard; they need a **partner.**

Your mission is to build that partner.

You are not just building a simulation; you are building a **Digital Co-Pil**
predicts its future, and communicates its needs in a way that empowers
intuitive human intelligence.

2. The Challenge: From Physics to Intelligence

To bring this Digital Co-Pilot to life, you must solve three fundamental

- **The Brain (Modeling):** You must translate the physical reality of
mathematical model.
- **The Life (Simulation):** You must breathe life into that model, cre
advancing simulation.
- **The Voice (Interaction):** You must build the bridge. You will crea
conversational intelligence—turning a machine that "runs" into a

3. Roadmap & Scope

The challenge is structured into three sequential (and cumulative) stage

Stage	Name	Focus
Stage 1	Teach your printer how it breaks	Modeling (The Logic E
Stage 2	Bring the printer to life	Digital Twin (The Simul
Stage 3	Let the printer talk	AI / Voice Interface (Th

4. Technical Constraints & Rules

To ensure a level playing field, all teams must adhere to the following c

- **Technology Stack:** There is no mandatory technology stack. Tea
cloud platforms.
- **Data Generation (Crucial): No pre-made telemetry or histori**

🛠️ Stage 1 — Modeling the Digital Twin

🎯 1.1 Stage Objective

To build the mathematical "brain" of the printer. You are creating a **Logic Engine** that simulates how components age, fail, and interact based on environmental and operational stressors.

In this stage, you are **not** building a UI or a simulation loop. You are building a **Mathematical Engine**. This engine must be able to take a set of environmental conditions as input and return the precise, updated state of every component in the printer. This engine will serve as the "source of truth" for the Digital Twin in all subsequent stages.

📋 1.2 System Overview & Component Map

Use the table below to plan your model. The minimum requirement to pass this stage is to model all **Mandatory** components.

These are some of the key components of the HP Metal Jet S100 that you can choose to model.

Subsystem	Component	Status	Primary Function
Recoating System	Recoater Blade	**Mandatory**	Spreads a thin, even layer of metal powder.
	Recoater Drive Motor	*Optional*	Drives the motion of the blade assembly.
	Linear Guide / Rail	*Optional*	Ensures smooth, straight-line movement.
Printhead Array	Nozzle Plate	**Mandatory**	Precisely ejects binder liquid onto the powder.
	Thermal Firing Resistors	*Optional*	Controls the temperature of the printhead.
	Cleaning Interface	*Optional*	Wipes the nozzle to prevent clogging.
Thermal Control	Heating Elements	**Mandatory**	Maintains the optimal build temperature.
	Temperature Sensors	*Optional*	Provides feedback on thermal stability.
	Insulation Panels	*Optional*	Reduces heat loss to the environment.

`Note`: You can add as many additional components as you wish. The more components you add, the more accurately the model will reflect real-world behavior. Make sure to clearly define each component's function and how it fits into the overall system.

If teams want to add extra components and are unsure which ones to include, they should ask Volunteer Support for guidance.

🚀 1.3 Component Modeling Guides (Physics Hints)

Since you are modeling a digital version of a physical machine, use these "physics hints" to guide your logic.

> **Importante**: Creo que falta incluir una explicacion de la printer, de los systemas y los componentes, si no, creo que va a ser demasiado complejo. LINK al video donde lo explica (Francho's video). Pequeño parrafo explicando como funciona

Subsystem A: Recoating System

- **The Recoater Blade:** Suffers from **Abrasive Wear**. As it moves through powder, it loses thickness and smoothness.
  - *Hint:* Consider how "Contamination" (an input vector) might accelerate this wear.
- **The Drive Motor & Rails (Optional):** Subject to **Mechanical Fatigue**.
  - *Hint:* High "Production Volume" should increase the probability of motor stall or rail misalignment.